ELARA and URITP: Project Brief Book

A plain-language guide for anyone, with no technical background needed.

1. The project in brief

Robots are everywhere: in factories, warehouses, hospitals, farms and homes. But every robot brand speaks its own private language. Learning to control one robot does not help you control another. Real robots are also expensive, so most students never get to try.

This project fixes both problems with two connected pieces:

- **URITP** is one common language that can be used to give instructions to any robot.
- **ELARA** is a free website where anyone can practise controlling robots, safely, in their own language.

The idea in one line: *Speak to any robot in your own language. Safe. Multilingual. Universal.*

2. The problem

- Every robot maker uses different commands, so skills do not carry over from one robot to the next.
- Real robots and powerful computers cost a lot, so students and schools are left out.
- Robot programming looks scary to beginners, and a mistake on a real robot can cause damage or injury.
- Most robot tools are in English only, which shuts out millions of learners.

3. The solution in simple words

Think of how a travel plug adapter works. Your device does not change, the adapter makes it fit any socket. URITP is like that adapter for robots. You write your instruction once, and URITP translates it into the exact commands each robot understands.

Right now, getting a robot to complete even one instruction means passing through many different languages and processes. Each robot brand has its own commands, and each step along the way has its own tools and rules. Someone has to know all of them and connect them by hand.

Our project aims to be the adapter that does all of that work. You give one instruction in your own language, URITP handles every step in between, and the robot gets the job done. You never need to learn the robot's own language.

This makes the adapter our main product, together with its security features. The safety checks, the safety limits and the emergency stop are not extras added on top. They are built into the adapter and run on every instruction before any robot moves.

ELARA is the playground built on top of it. It runs inside a web browser, so there is nothing to install and no special equipment is needed. It works on a phone or a laptop.

4. What is URITP?

URITP stands for **Universal Robot Instruction/Translation Protocol**. It is a language and a set of rules that work for any kind of robot: a wheeled delivery robot, a factory arm, a drone, a four-legged robot dog or a humanoid.

Its main ideas are easy to remember:

- **Say the goal, not the motion.** You say where something should end up. The robot works out how to get there.
- **Check before moving.** The program is checked before any motor turns. If a robot cannot do something, you are told why instead of the robot guessing.
- **Easy to read.** A school student should be able to read a program and understand it.
- **Units are always written.** You write 10cm or 0.2m/s, never a bare number, so nobody confuses centimetres with metres.
- **Safety cannot be switched off.** Every program runs inside safety limits that code cannot remove.
- **Open to every maker.** Robot makers can add their own special features without breaking the common rules.

A tiny example of what URITP looks like:

```
robot rover = connect("office:rover-2")
move rover.base to places.mailroom
disp "Please place the parcel on my tray"
move rover.base to places.dock
```

Even without knowing code, you can guess that the rover goes to the mailroom, asks for a parcel and then returns to its dock.

5. What is ELARA?

ELARA is a free 3D robotics simulator that runs in the browser. It turns everyday speech into URITP instructions, then lets you watch a virtual robot carry them out.

What it offers:

- **Speak naturally.** You can type or say something like "laal gend uthao aur tokri mein rakho" (pick up the red ball and put it in the basket). Supported languages include English, Hindi, Hinglish, Spanish and Arabic.
- **Two 3D worlds.** The Kitchen has a robot arm. The Town has a rover. There are 11 guided missions plus free play.
- **A built-in safety guard called Rakshak.** If an instruction would break a safety rule, such as entering a no-go zone or going too fast, Rakshak stops it, highlights the problem line and explains in simple terms what went wrong.
- **Easy sharing.** Everything runs on the user's own device, with no login needed. Users can share their work through a link, short clips and completion certificates.

6. How it works, step by step

1. **You speak or type** a request in your own language.
2. **An AI layer translates** it into URITP.
3. **The checker reviews it.** Does the robot have this ability? Are the units right? Are the safety limits respected?
4. **If it is safe and possible, the robot does it.** If not, it is stopped before moving and you get a clear reason.
5. **You get the result.** The robot reports success, failure or cancellation in the same way, whatever the robot.

A spoken request can never make a robot do anything that a carefully written program could not do.

7. Safety, the heart of the project

- Programs are checked **before** the robot moves.
- Robots never quietly do a "close enough" version of your request. They either do exactly what was asked or refuse and explain why.
- Every robot has an emergency stop that works at any moment.
- Every robot keeps its own safety limits, even if a program tries to break them.
- Error messages say what went wrong, where, and what to try instead. They can be shown in the user's own language.

8. One language, many robots

Robot	What it does with URITP
Wheeled rover	Delivers a parcel across an office
Robot arm	Picks up a cup and places it on a tray
Drone	Flies over a roof, takes photos and lands, returning home if its battery runs low
Robot dog	Patrols a factory at night and raises an alert if something is too hot
Humanoid	Pours tea in a kitchen using both hands
Welding arm	Welds along a seam, using the maker's special feature if available

9. Who is it for?

- **Everyday users and students** speak to robots in their own language.
- **Developers and engineers** write URITP directly when they need precision.
- **Robot makers** add URITP support by following the official guidebook, and keep their own unique features.
- **Schools and training institutes** use it to teach robotics without buying robots.

10. The five offerings (MSPs)

The project grows through five offerings, each one leading naturally to the next.

#	Offering	Who it is for	What they get
1	**Free Playground**	School students (Classes 8 to 12), kids and beginners	A completely free browser simulator with multilingual input, the Rakshak safety guard, 11 missions and shareable certificates. Its job is to attract very large numbers of users at almost no cost.

#	Offering	Who it is for	What they get
2	**URITP for Schools and STEM Labs**	Schools, coaching institutes and lab teachers	A yearly school licence or teacher subscription with progress dashboards, automatic mission grading, co-branded certificates and ready-made lesson plans.
3	**Powered by Pi and Arduino Maker Packs**	Hobbyists, robotics clubs and makers	Extension keys and hardware packs so code tested in the browser can be safely moved onto real Arduino and Raspberry Pi builds.
4	**ROS 2 Advanced Academy**	Senior school students (Classes 11 and 12), engineering aspirants and ROS 2 developers	A paid tier with advanced simulation and safety-checking, linked to the Open Source Robotics Alliance, to earn real robotics credentials.
5	**Enterprise Pilot and Industrial Certification**	Vocational institutes, technical universities and industrial partners	Paid pilots and safety certification modules aligned with upcoming EU Machinery Regulations, plus a path into the Universal Robots (UR+) partner programme and safety audit logs.

11. How the business grows

The model is often called **sandbox to factory**:

1. **Free for everyone.** With no login walls and no server costs, students anywhere can try it and share it. This brings in users by word of mouth.
2. **Partners spread it.** In India, the plan is to work with PhysicsWallah (PW) on a robotics module and a co-branded PW and URITP certificate. Worldwide, the plan is to reach classrooms through Raspberry Pi resellers, the Arduino community and the ROS 2 educational network.

3. **Institutions pay.** Schools, labs and companies pay for dashboards, hardware kits, advanced tiers and industrial certification.

More ways to earn

Three more income streams are added as the project grows. They are paid for by robot makers, companies and sponsors, and the free playground stays free.

- **Paid certification for robot makers.** The guidebook already lists the rules a robot must follow to be called URITP-compatible. A maker pays to have its robot tested against those rules and, once it passes, to use the official "URITP-compatible" name and logo. This is similar to how makers of phones and Wi-Fi devices pay to carry a trusted label.
- **Free core, paid extras for companies.** The language and the basic safety checker stay free for everyone. Companies pay for the extras that matter to them: dashboards to watch many robots, safety records for audits, custom no-go zones, support and private installation.
- **Sponsored challenges and certification exams.** Robot and hardware companies sponsor robotics challenges and competitions to reach students and makers. Learners who want a recognised proof of skill can take an optional certification exam, which may carry a small fee.

In short, students are never charged, and revenue comes from the institutions and companies that benefit from students who are ready for real robotics.

12. Why it matters

- It opens robotics to learners who could never afford a robot.
- It breaks the language barrier, so a student can learn in Hindi, Arabic, Spanish or English.
- It builds safe habits from the first day.
- It gives robot makers a shared standard instead of many private ones.

13. Small glossary

- **Robot:** a machine that can move and act, such as a rover, arm, drone, robot dog or humanoid.
- **Simulator:** software that shows a virtual robot, so nothing real can break.
- **Protocol:** an agreed set of rules for two sides to understand each other.
- **URITP:** the common language and rulebook for robots.
- **ELARA:** the free playground where people practise URITP.

- **Rakshak:** ELARA's built-in safety guard (the word means protector).
- **Checker:** the tool that reviews a program before any robot moves.
- **Manifest:** a robot's list of what it can and cannot do.
- **Extension:** a special feature added by one robot maker, clearly labelled with the maker's name.
- **ROS 2:** a widely used open-source toolkit for robotics.
- **Raspberry Pi and Arduino:** small, low-cost computer boards popular with students and makers.
- **MSP:** one of the five offerings described in section 10.

14. Quick questions and answers

Do I need to know coding? No. You can speak or type in your own language, and the system writes the code.

Do I need a real robot? No. ELARA shows virtual robots in your browser.

Is it free? Yes for students. Paid plans are for schools, labs, advanced learners and companies.

Is it safe? Programs are checked before anything moves, and safety limits cannot be switched off.

Can a robot maker use it? Yes. The official guidebook explains how, and makers can keep their special features through extensions.